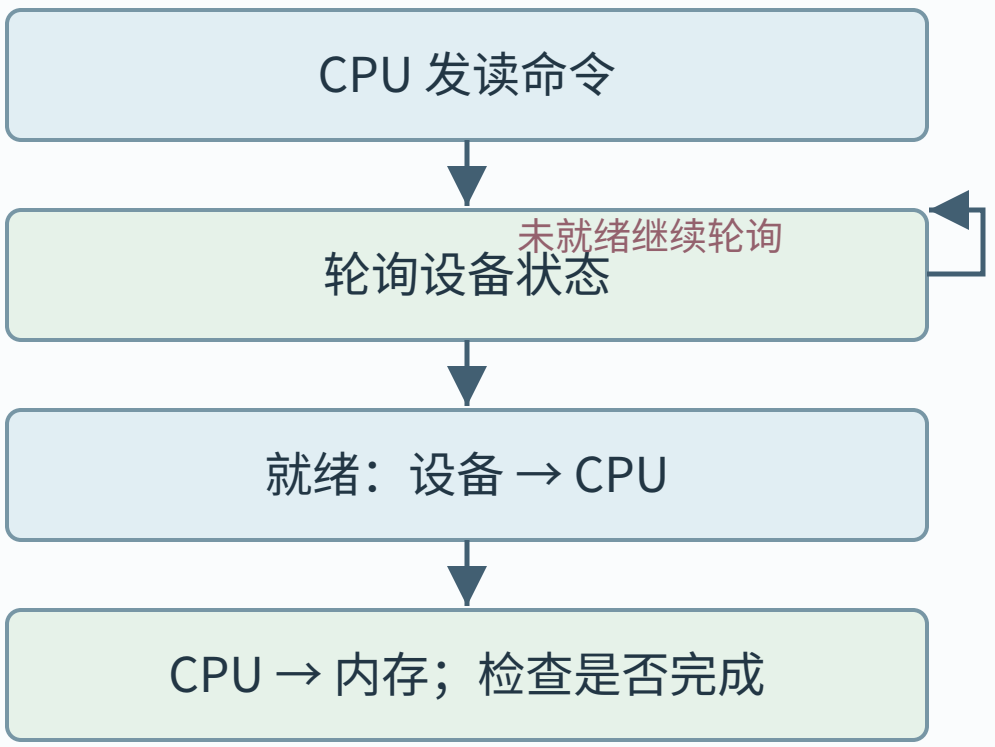
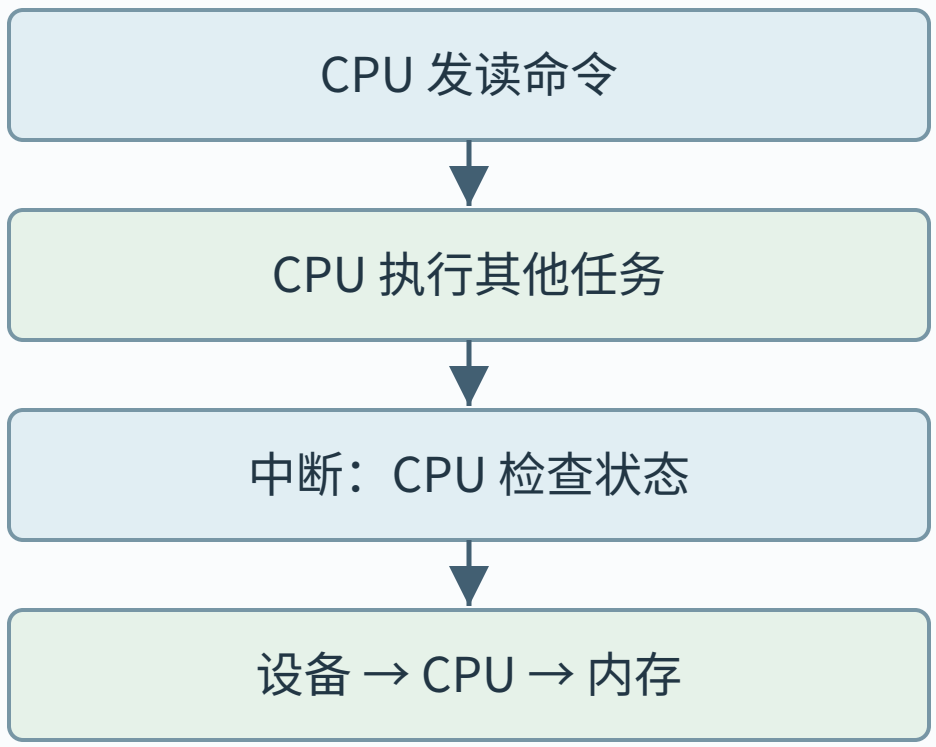
① 程序直接控制

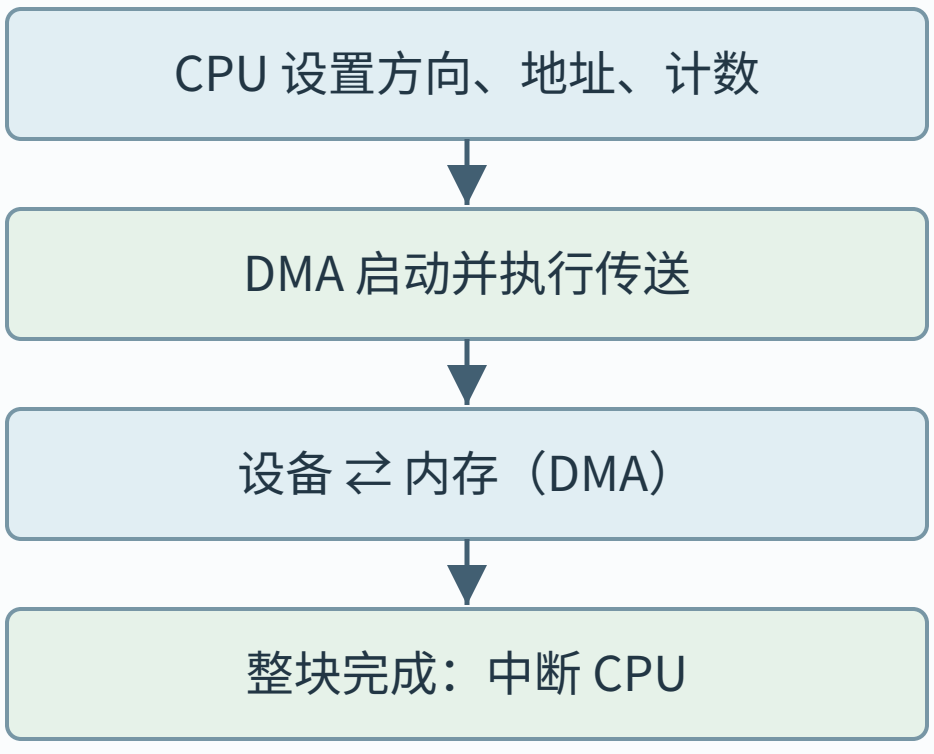


② 中断驱动



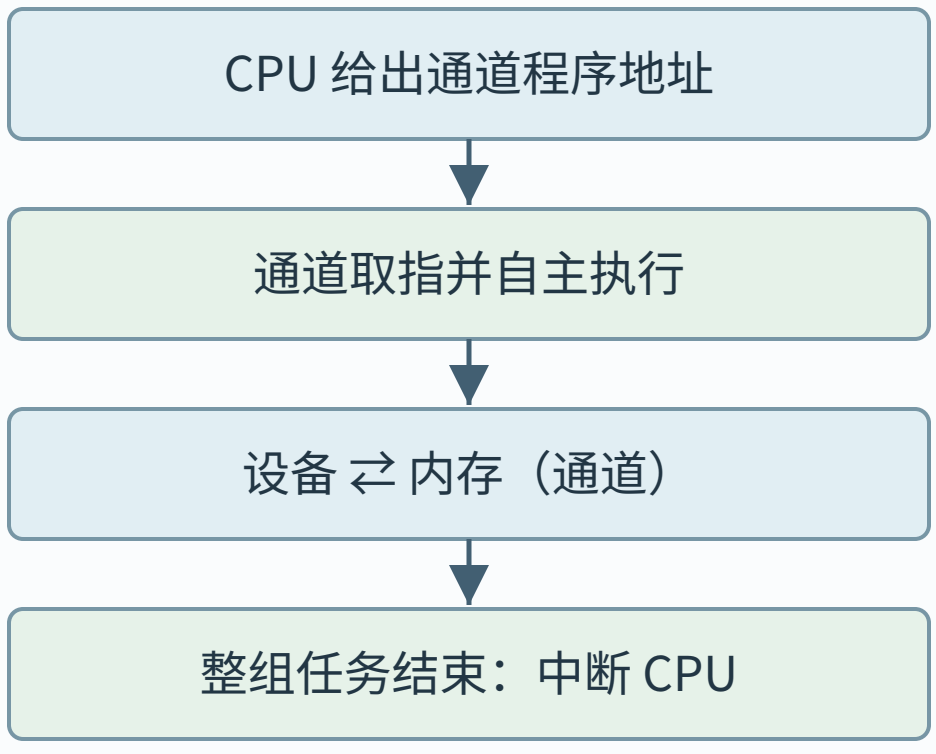
CPU 在每字中断服务中搬运数据

③ DMA 控制



CPU 介入启动、异常和结束处理

④ 通道控制



通道程序位于内存，组织一组数据块